OpenClaw OLED Display Tutorial

> Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-OpenClaw Prompts], [05-AI Voice Interaction Configuration], and [06-Three Conversation Scheme Configuration and Testing] in the `Preparation for Use` directory. If you are currently only using TUI / WhatsApp, you can skip `05` for now. This document assumes the basic environment is already prepared and only expands on OLED-related content.

1. Tutorial Objectives

This tutorial demonstrates OpenClaw's natural language control capabilities through OLED peripherals. After completing this tutorial, readers will be able to:

- Understand the basic idea of OpenClaw controlling OLED displays
- Interact with OpenClaw via three methods: WhatsApp, TUI, and voice code
- Complete four OLED cases: text display, metric display, weather card, animation display
- Understand the complete pipeline from "user input" to "screen display"

2. Hardware Preparation

- OpenClaw main board
- OLED display `[Default SSD1306 128x32 / I2C Address 0x3C]`
- `[Optional] Microphone module`
- Wiring diagram

3. Software Preparation

Code entry points used in the tutorial

- Low-level driver code: `/home/sunrise/hardware_hal_demo/`
- OLED control command: `/home/sunrise/hardware_hal_demo/oled`
- Temperature and humidity linkage display: `/home/sunrise/hardware_hal_demo/dht_oled`
- Voice entry point: `/home/sunrise/voice_to_openclaw/src/main.py`

The `oled` tool already supports the following capabilities:

- Display text `text`
- Display image `image`
- Display animation `animation`
- Display weather card `weather`

- Clear screen `clear`

If you just want to quickly display temperature and humidity readings on the OLED, you can also use `dht_oled` directly.

```
dht_oled  # Display temperature and humidity data to OLED
oled text --text 'Hello, World' # Display text
oled weather --location 'Shenzhen' --condition 'Sunny' --temperature '26°C' --
humidity '57%' # Weather card
oled image --path ~/hardware_hal_demo/assets/oled/mountain_photo_128x32.png #
Display image
oled animation --path  ~/hardware_hal_demo/assets/oled/sunrise_anim_128x32.gif #
Display GIF animation
oled clear   # Clear screen
```

Note: For GIF and image display, you can ask AI to generate images with 128x32 resolution and place them in this path: ~/hardware_hal_demo/assets/oled/

For more commands, you can check the md documents in this folder:

```
/home/sunrise/hardware_hal_demo/README.md
```

If the low-level commands can be executed normally, it means the hardware connection and communication with the RDK X5 motherboard are normal.

4. Testing OLED Display

Before starting this tutorial, please complete a basic `openclaw tui` conversation self-check first; if you haven't done so, see [06-Three Conversation Scheme Configuration and Testing]. After confirming that OpenClaw can respond normally, proceed to the OLED-specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```

```
OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws—type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

You can try entering the following content in the terminal:

- Display "Hello, OpenClaw" on the OLED
- Display "Start learning OLED today" on the screen
- Clear the OLED screen
- Display current status "System OK" on the OLED

```
Display "Hi, OpenClaw" on OLED

"Hi, OpenClaw" is now displayed on the OLED screen. ✅

Display "Today we start learning OLED" on the screen

"Today we start learning OLED" is now displayed on the screen. The text was
automatically wrapped into two lines to fit the OLED display. ✅

Clear the OLED screen

OLED screen cleared. ✅
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
46k/1.0m (5%)
```

5. Three Schemes for Conversing with OpenClaw

Scheme	Suitable Scenarios	Advantages	Suggested Positioning
WhatsApp	Remote control, classroom demonstration	Easy access, suitable for multi-user experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	Get the control pipeline running first
Voice Code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration scheme

If you have selected the AI voice module + speaker, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

- Modify whether to enable broadcast parameter

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether to broadcast. After modifying and saving, run the following program in the terminal.
```

- Enter in terminal:

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

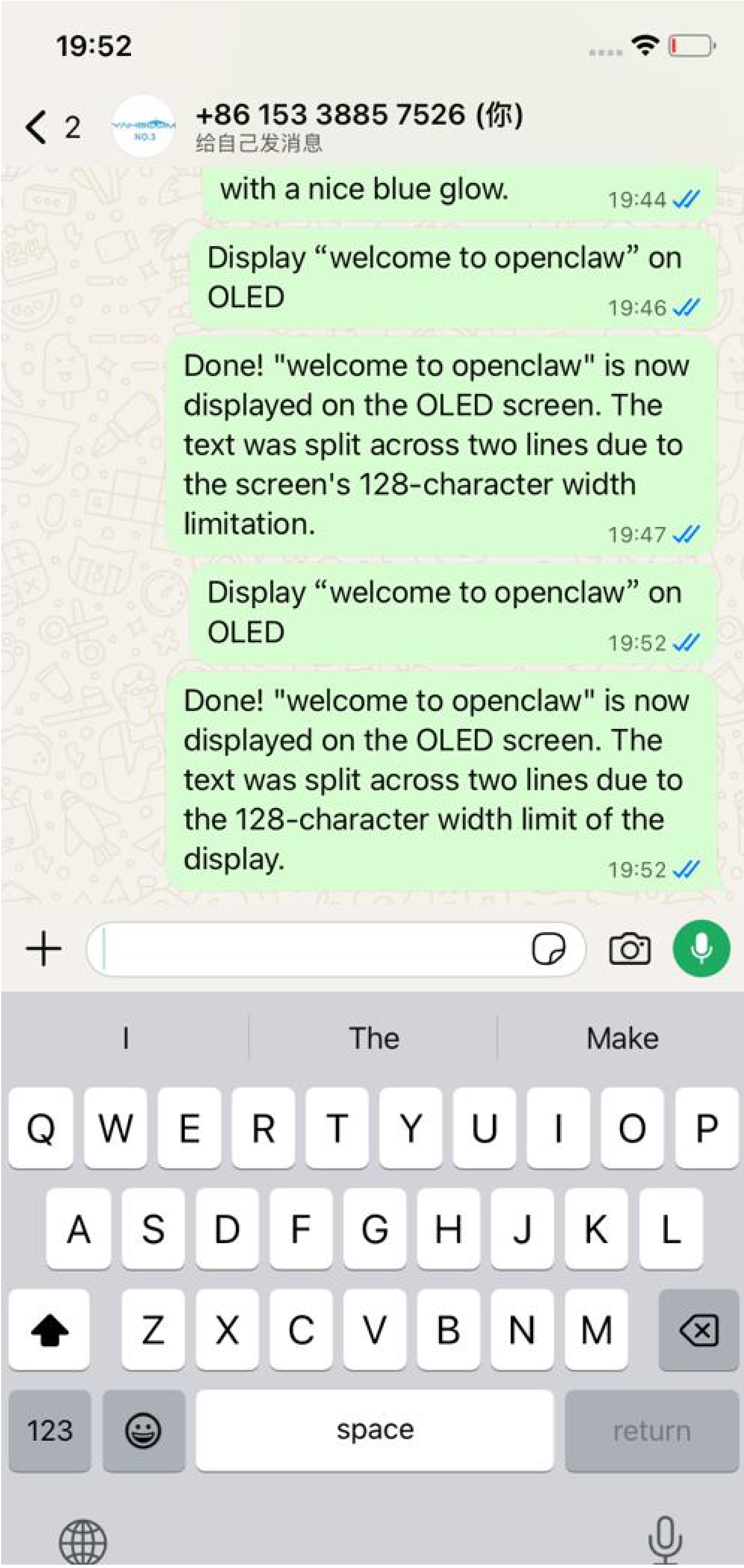
```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

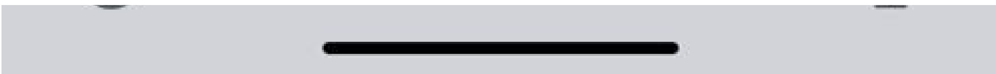
5.1 WhatsApp Scheme

For WhatsApp access steps, please refer directly to [06-Three Conversation Scheme Configuration and Testing]. This section only retains conversation examples suitable for the OLED tutorial.

After creation, you can converse directly, for example:

- Display "Welcome to OpenClaw" on the OLED
- Help me display current temperature and humidity data on the OLED
- Clear the OLED screen

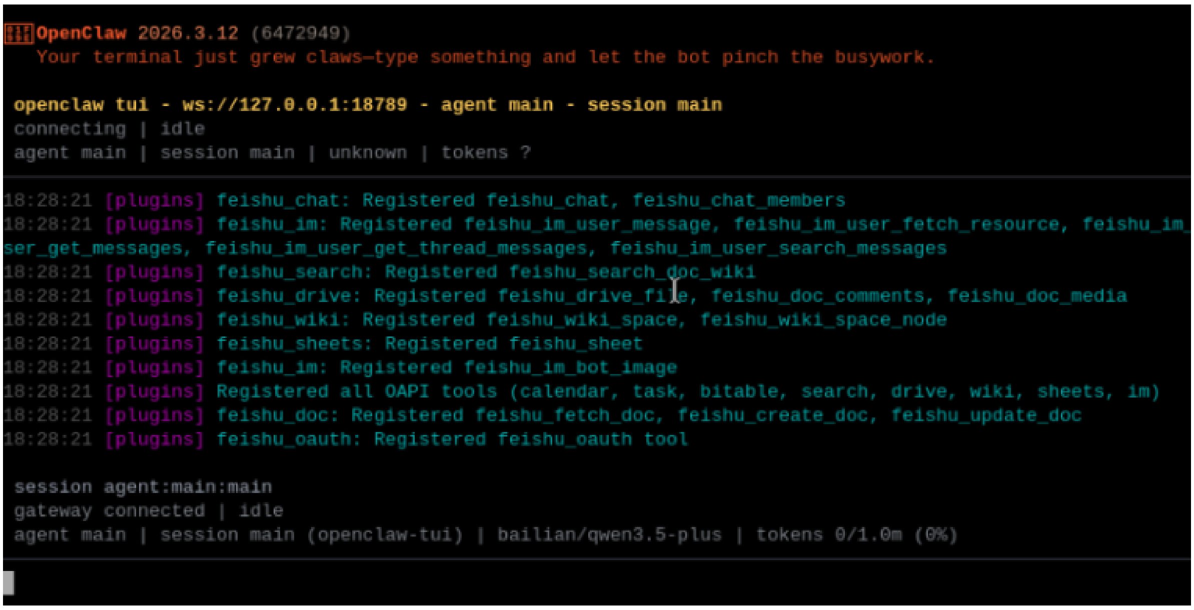

19:52
2
+86 153 3885 7526 (你)
给自己发消息
with a nice blue glow. 19:44
Display “welcome to openclaw” on OLED 19:46
Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the screen's 128-character width limitation. 19:47
Display “welcome to openclaw” on OLED 19:52
Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the 128-character width limit of the display. 19:52
I
The
Make
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
123
space
return



5.2 TUI Scheme

If you have completed the TUI self-check in Part 6, you can continue the conversation by entering the following command in the terminal:

```
openclaw tui
```



5.3 Voice Code Scheme

```
Requires selection of AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration].

```
cd /home/sunrise/voice_to_openclaw
source .venv/bin/activate
python -m src.main
```

After wake-up, you can directly speak control content, for example:

- Display "Hello" on the OLED
- Change the screen content to "Welcome to OpenClaw"
- Clear the screen

```
2026-05-14 17:25:01 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串
口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说"清空上下文"可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 8520 ms 语音, 正在转写

[你] Display high on old.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! "high" is now displayed on the OLED screen.
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Clear the screen.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! The OLED screen has been cleared.
```

6. Comprehensive Cases

Note: The following are conversation demonstrations conducted via `openclaw tui`. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: Text Display

Experiment Objective

Let the OLED display a fixed segment of text to complete the most basic screen control.

Effect Description

After receiving the instruction, OpenClaw will render the specified text on the OLED screen. This case is suitable for verifying whether text rendering, line breaks, and screen refreshing are normal.

Example Instructions

- Display "Hello, OpenClaw" on the OLED
- Display "Today's task is to test OLED"
- Change the screen content to "System Ready"

6.2 Case 2: Metric Display

Experiment Objective

Let the OLED display single-line status metrics in a more compact way.

Effect Description

This mode is suitable for compressing short information such as temperature, humidity, battery level, and status values into one line for display, more like a simple dashboard reading.

Example Instructions

- Display "Temp 26C Hum 57%" on the OLED
- Write the current status as a short single-line display
- Change the screen content to "CPU OK RAM OK"

6.3 Case 3: Weather Card

Experiment Objective

Let the OLED display a more complete weather information card.

Effect Description

The screen can display information such as location, weather, temperature, and humidity. This case is suitable for making weather stations, environmental monitoring, or information panels.

Example Instructions

- Display Shenzhen sunny 26 degrees humidity 57% on the OLED
- Make a concise weather card
- Format the current temperature and humidity as a card style for display

6.4 Case 4: Animation Display

Experiment Objective

Let the OLED play images or animations to showcase richer visual effects.

Effect Description

In addition to static text, the OLED can also display bitmaps, GIF animations, or simple self-check animations. This case is very suitable for making boot pages, expression animations, or classroom demonstrations (you can prepare photos or GIF animations to be displayed and place them in the path /home/sunrise/hardware_hal_demo/assets/oled).

Example Instructions

- Play a simple animation on the OLED
- Display an icon image
- Make a boot welcome animation